

Ayush Kumar Singh

Senior Data Engineer

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Professional Summary

Senior Data Engineer with 3.5+ years building real-time and batch data infrastructure for high-volume fintech platforms. Expert in CDC, Apache Kafka, AWS lake house (Hudi, Iceberg), PII/DPDP compliance, and end-to-end ETL pipelines. Proven delivery across streaming systems, data governance, observability stacks, and AI-driven pipeline automation. Experienced in building GenAI-powered data agents (Microsoft Fabric, Copilot) with semantic modeling and row-level security Proficient in Python, PySpark, and a broad AWS services ecosystem.

Education

- Symbiosis Institute of Technology August 2019 - June 2023
B. Tech in Computer Science and Engineering | 7.59 CGPA – Honours in Data Science

Work Experience

- Bajaj Broking | Senior Data Engineer Aug 2025 - Present
- Built and published an AI-grounded Copilot agent in Microsoft Fabric on a Power BI semantic model of client feature data, with row-level security restricting each Relationship Manager to their own clients, enabling self-serve natural language Q&A via Microsoft Teams.
 - Architected and maintained scalable real-time data pipelines by implementing Change Data Capture (CDC) on MS SQL Server and streaming events through Kafka, reducing transactional database query load by 40–60% and improving data availability for downstream systems.
 - Developed and deployed an AI-powered agent on AWS to monitor ETL pipelines, automatically detecting failures, data anomalies, and performance bottlenecks to improve pipeline reliability.

- Bajaj Broking | Data Engineer Jul 2023 – Jul 2025
- Architected scalable AWS data lake infrastructure using Hudi, Iceberg, and Parquet formats; applied encryption and governance controls to ensure PII compliance and enable secure, high-performance data access for analytics teams.
 - Engineered a comprehensive monitoring solution for a trading application, incorporating Grafana, Fluentd, and Elasticsearch. Integrated real-time logs and on-premises databases, cutting report creation time by 80% and improving API performance by 40%.
 - Worked with cross-functional teams to design reporting data models, power real-time dashboards, and support AI-driven workflows by transforming raw data into structured, governed datasets for analytics and decision-making.
 - Developed a robust ETL framework for a Customer Data Platform (CDP) project using AWS services (Glue, S3, Athena). Enhanced data integration and processing workflows, ensuring compliance with data security standards by encrypting and managing Personally Identifiable Information (PII).
 - Led the integration of vendor data pipelines into the enterprise data lake house, designing scalable ingestion workflows that transformed external datasets into governed, analytics-ready tables for downstream reporting, analytics, and ML workloads.

- Bajaj Broking | Data Engineer Intern Jan 2023 – Jun 2023
- Built a Data Discovery solution for identifying and classifying PII across multiple databases, including names, phone numbers, emails, Aadhaar, and PAN. Implemented encryption in compliance with Data Protection and Digital Privacy (DPDP) regulations.
 - Designed and implemented real-time engagement systems for targeted campaigns, increasing campaign CTR by 18–35% and improving client response rates by 33%.
 - Engineered an Aadhaar masking solution using Easy OCR and Python, improving data masking accuracy from 70% to over 95%.
 - Executed RFM segmentation models on trading data to power customer lifecycle analytics and targeted campaigns, driving a 12% increase in customer retention.

Technical Skills

Languages: Python, PySpark, Go, Fluentd, R.
Database: MSSQL, MySQL, PostgreSQL, MongoDB, Elasticsearch, Redis, Amazon Athena.
Tools: AWS, Git, NGINX, Grafana, AWS Services (S3, Glue, Lambda, API Gateway, SNS, Athena, KMS, IAM, Event Bridge).
Machine Learning & AI: TensorFlow, PyTorch, Scikit-learn, Keras, XGBoost, Deep Learning, AI Agents, Fabric Agent.
Data Engineering & Streaming: Apache Kafka, Change Data Capture (CDC), Real-time Streaming Pipelines, Batch Data Pipelines, Data Modeling.
Technologies/ Frameworks/ Libraries: Apache HUDI, Apache Iceberg, Apache Spark, Jupyter Notebooks, Docker, Parquet.
Soft Skills: Cross-functional Collaboration, Technical Problem Solving, Stakeholder Communication, Technical Leadership.

Projects

- AR/ Augmented Reality Quiz | Python, Flutter, Media Pipe Face mesh
- Innovated an Augmented Reality Quiz application using Python, Flutter, and Media Pipe Face mesh. The app leverages face-tracking technology to lock options based on the user's facial movements, enhancing interactive and immersive quiz experiences.

Achievements

- Super Heroes Award | Bajaj Broking
- Recognised for consistently exceeding expectations and delivering exceptional value, demonstrating outstanding performance and dedication.